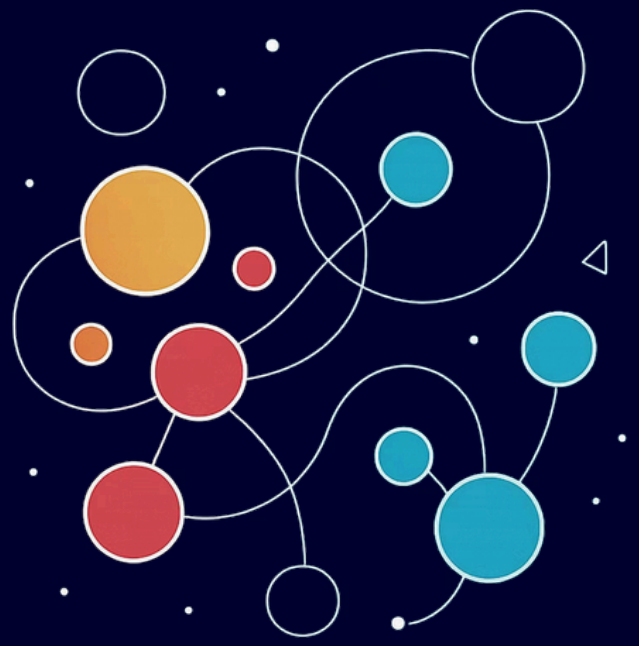
Cluster Analysis in Data Mining

Understanding Unsupervised Learning Through Grouping Patterns



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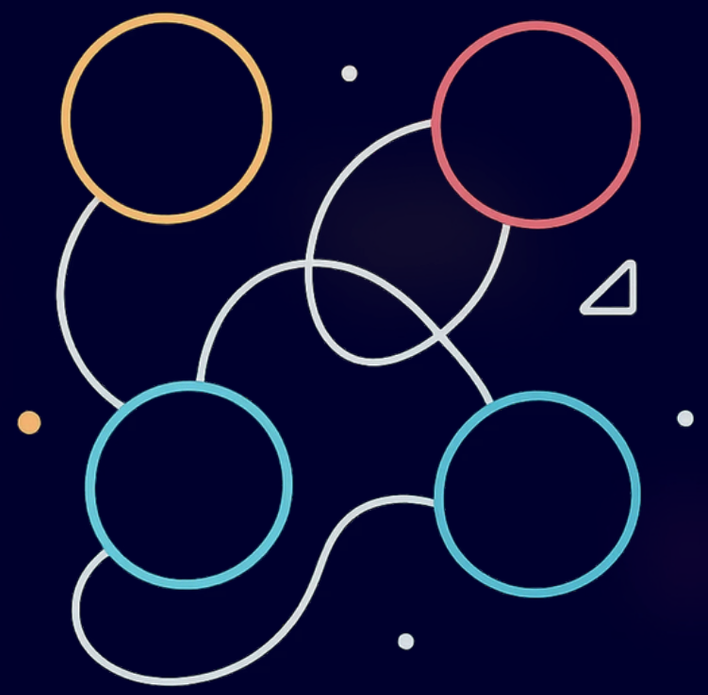
CSE-CORE-I

What is Clustering?

Clustering is an**unsupervised learning technique** that groups similar data points together while keeping dissimilar points separate. Unlike classification, clustering works without predefined labels, discovering hidden patterns and structures in data.

The goal is to maximize intra-cluster similarity (points within the same cluster are alike) while maximizing inter-cluster dissimilarity (different clusters are distinct from each other).



Applications

Customer segmentation, image analysis, document clustering, anomaly detection

Key Benefit

Reveals hidden patterns without requiring labeled training data

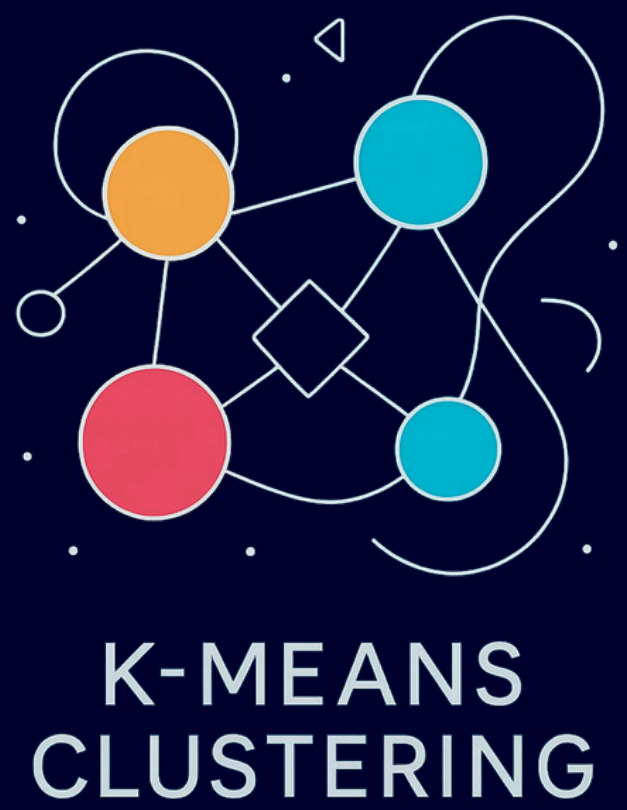
Challenge

Determining the optimal number of clusters and evaluating quality

Partitioning Methods: K-Means Clustering

K-means is one of the most popularpartitioning algorithmsthat divides datainto **k**

clusters by minimizing the within-cluster sumof squares.Thealgorithm iteratively assigns points to the nearest centroid and recalculates centroids until convergence.



01

Init ialize

Select k initial centroids (randomly or using methods like k-means++)

02

Assign

Assign each data point to the nearest centroid

03

Update

Recalculate centroids as the mean of all points in the cluster

04

Repeat

Continue until centroids stabilize or maximum iterations reached

Advantages

- Simple and easy to implement
- Computationally efficient for large datasets
- Works well with spherical clusters

Limitat ions

- Requires specifying k in advance
- Sensitive to initial centroid placement
- Struggles with non-spherical shapes
- Outliers can significantly affect results

Hierarchical Clustering: Building Trees of Clusters

Hierarchical clustering creates a tree-like structure (dendrogram) showing relationships between clusters at different scales. This approach doesn't require

specifying the number of clusters upfront and provides insights into data structure at multiple granularities.

1

Agglomerative (Bottom-Up)

Start with each point as its own cluster, then repeatedly merge the closest pairs until one cluster remains

2

Divisive (Top-Down)

Start with all points in one cluster, then recursively split clusters until each point is separate

Linkage Criteria

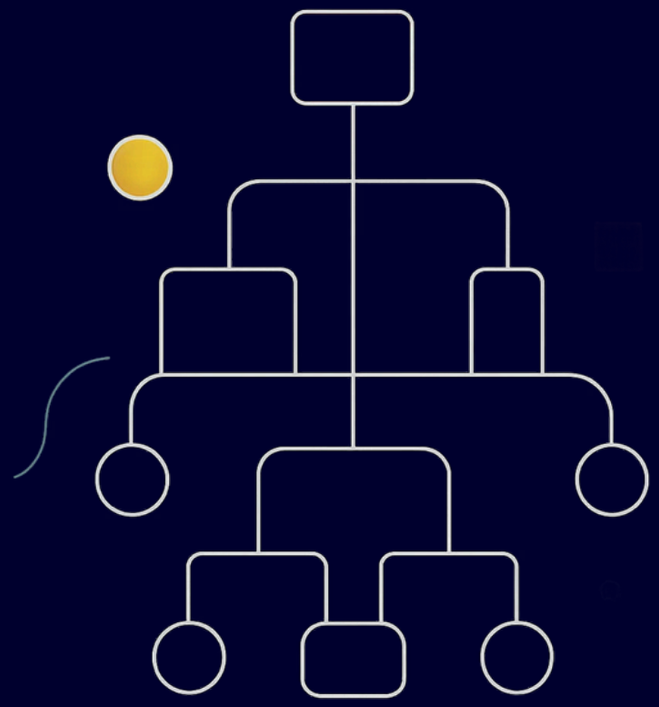
Single (minimum distance), complete (maximum distance), average (mean distance), or centroid-based

Strengt hs

Produces interpretable dendrograms, no need to specify k, reveals multi-scale structure

Weaknesses

Computationally expensive for large datasets, cannot undo previous merges/splits



Probabilistic Model-Based Clustering

Model-based clusteringassumes data is generated froma mixtureof probability distributions. Each cluster corresponds to one distribution component, and the algorithm estimates parameters using techniques like Expectation-Maximization (EM).

The most common approach uses **Gaussian Mixture Models (GMM)**, where each cluster follows a multivariate normal distribution. GMM provides soft clustering through membership probabilities rather than hard assignments.

Flexible Shapes

Can model clusters with different sizes, shapes, and orientations through covariance matrices

Soft Clustering

Provides probability of membership for each point, capturing uncertainty in assignments

Statistical Foundation

Uses rigorous probability theory with clear assumptions and interpretability

Key Limitation: Assumes data follows specific distribution family, which may not match real-world data characteristics

BIRCH: Balanced Iterative Reducing and Clustering

BIRCH is designed forverylargedatasets thatcannot fitinmemory.ItbuildsacompactsummarydatastructurecalledaClusteringFeature Tree (CF Tree) that captures clustering information while using limited memory.

Build CF Tree

Scan data and incrementally build height-balanced tree where each node stores clustering features (count, linear sum, squared sum)

Cluster Subtree

Apply clustering algorithm to leaf nodes of CF Tree to produce initial clusters

Refine Clusters

Optional step to improve quality by redistributing points or removing outliers

Strengt hs

- Handles massive datasets efficiently
- Single scan through data
- Incremental clustering capability
- Memory-efficient CF Tree

Weaknesses

- Requires threshold parameter tuning
- Assumes spherical clusters
- Quality depends on CF Tree structure
- Struggles with high-dimensional data

DBSCAN: Density-Based Spatial Clustering

DBSCAN identifies clustersas denseregionsseparatedbysparseareas.Itexcelsatfindingarbitrarilyshapedclusters and detecting outliers, making it robust to noise and effective for spatial data.

Core Points

Points with at least MinPts neighbors within radius ·

Border Points

Points within · of a core point but with fewer than MinPts neighbors

Noise Points

Points that are neither core nor border points

Arbitrary Shapes

Finds clusters of any shape, not limited to spherical patterns

Noise Handling

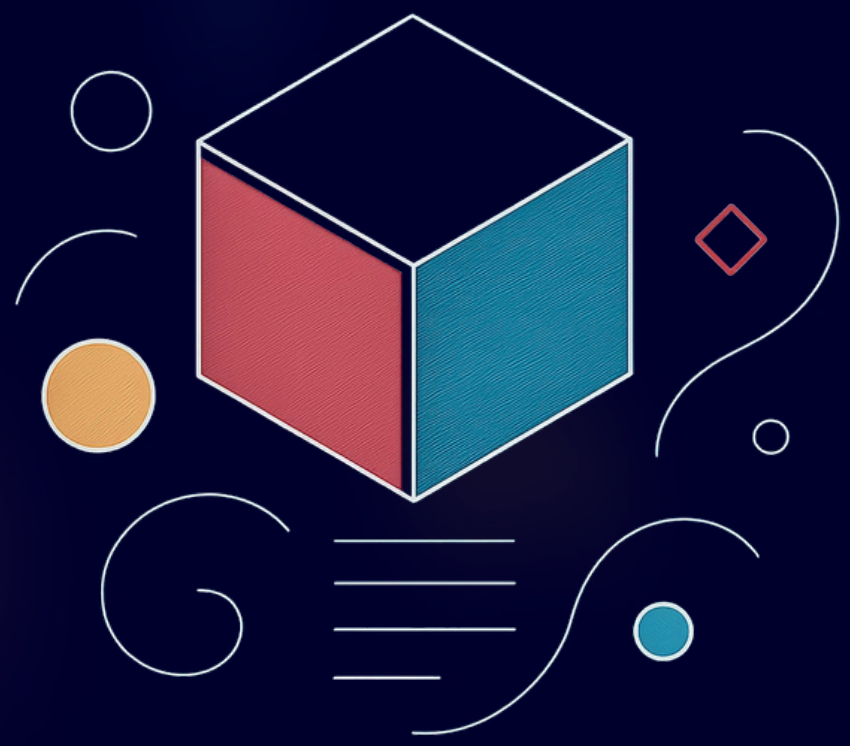
Naturally identifies and separates outliers from clusters

Parameter-Free (Almost)

Doesn't require specifying number of clusters, though · and MinPts need tuning

Challenge: Performance degrades in high-dimensional spaces due to the "curse of dimensionality" where density becomes less meaningful

Grid-Based Methods: STING & CLIQUE



STING: Statistical Information Grid

Partitions spaceinto rectangular gridcellsat multiple resolutions. Each cell stores statistical information (count, mean, standard deviation). Clustering queries use this precomputed grid to efficiently find regions with similar statistical properties.

- Fast query processing with O(1) time complexity
- Multi-resolution analysis capability
- Statistical measures enable significance testing

Common STING Parameters

- Gridresolution levels
- Threshold for dense cells
- Statistical significance level

CLIQUE: Grid-Subspace Method

Combines grid-basedand subspaceclustering for high-dimensional data. Partitions each dimension into intervals, then identifies dense units and subspaces where clusters exist. Automatically discovers relevant subspaces for each cluster.

- Handles high-dimensional data effectively
- Discovers clusters in subspaces
- Automatic dimensionality reduction

Common CLIQUE Parameters

- Number of intervals perdimension
- Threshold for dense units
- Subspace coverage requirement

Clustering Algorithm Comparison

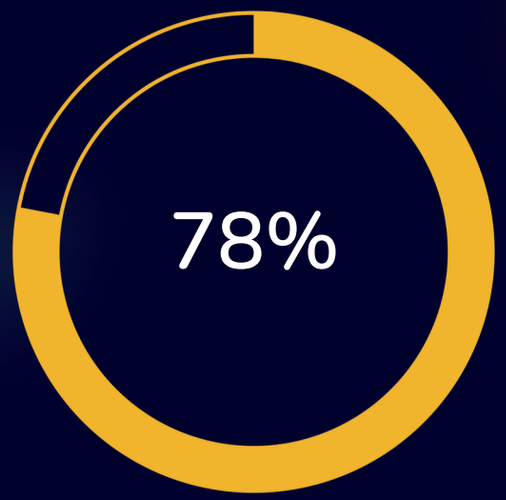
Eachalgorithmhasdistinctcharacteristicssuitedtodifferentdatascenarios.Understanding their trade-offs helps select the right approach for your specific problem.

Method	Best For	Cluster Shape	Parameters	Time Complexity
K-Means	Spherical, well-separated clusters	Spherical only	k, max iterations	O(nkdi)
Hierarchical	Multi-scale analysis, interpretability	Any shape	Linkage, distance	$O(n^3)$
GMM	Probabilistic interpretation	Ellipsoidal	k, covariance type	$O(nkd^2)$
BIRCH	Large datasets, limited memory	Spherical	Threshold, branching	O(n)
DBSCAN	Arbitrary shapes, noise detection	Any shape	·, MinPts	O(n log n)
STING	Spatial data, fast queries	Grid-defined	Resolution, threshold	O(n)
CLIQUE	High-dimensional subspaces	Subspace-based	Intervals, threshold	O(2^d n)

Note: n = number of points, k = number of clusters, d = dimensions, i = iterations. Actual performance varies with implementation and data characteristics.

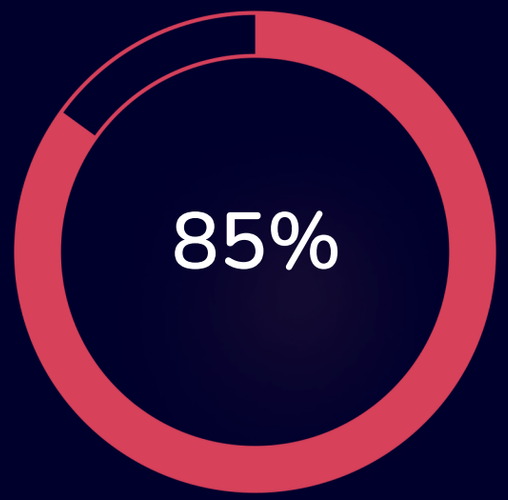
Evaluating Clustering Quality

Sinceclusteringisunsupervised, evaluationrequires indirectmeasures.Weuse **internal metrics** (using only data), **external metrics** (with ground truth), and **relative metrics** (comparing different clustering results).



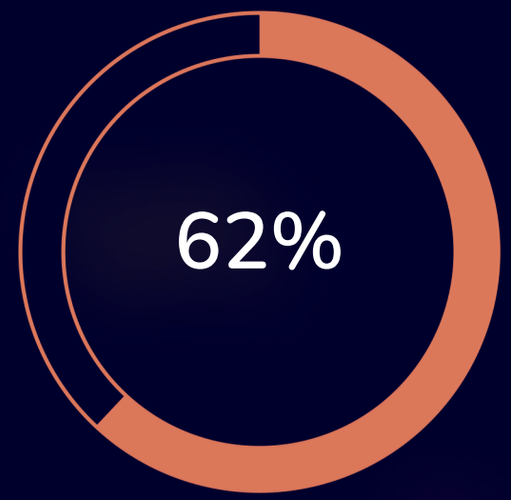
Silhouette Coefficient

Measures how similar points are to their own cluster versus others. Ranges -1 to 1, higher is better.



Within-Cluster Sum of Squares

Measures compactness by summing squared distances to centroids. Lower values indicate tighter clusters.



Calinski-Harabasz Index

Ratio of between-cluster to within-cluster variance. Higher values indicate better separation.

External Validation

When true labels exist: Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), Fowlkes-Mallows Index

Stability Testing

Compare results across subsamples or algorithm runs. Consistent clusters indicate robust findings

Visual Inspection

Plot clusters using dimensionality reduction (PCA, t-SNE). Human judgment identifies meaningful patterns